

Movement

ORGANISMS HAVE DEVELOPED DIFFERENT WAYS OF MOVING.

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Organisms move by changing the shape of their body to propel themselves forward. In complex animals these body changes are controlled by muscles, bundles of protein that exert pulling forces on body parts.

Modes of locomotion

Animals move in order to find food, escape a threat, or locate mates. The precise mode of locomotion (movement) used depends heavily on their habitat. Plants and fungi cannot move in the same way—their stiff cell walls make their bodies too rigid. However, many single-celled organisms, such as most protists and algae, can move by using extensions called flagella or cilia in the search for food or better conditions.



△ Flying
Wings are modified limbs that create lift and thrust forces to carry birds, bats, and some insects through the air.



△ Swinging
Tree-dwellers require a large decision-making brain and nimble limbs to control climbing and jumping.



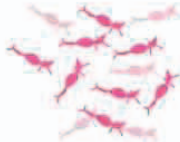
△ Walking
Most land animals walk on four legs (quadrupedal), although humans and flightless birds walk on two (bipedal).



△ Burrowing
Burrowers have powerful limbs for digging or are slender enough to be able to wriggle through soft soils.



△ Floating
The Portuguese man-of-war cannot move itself, but it is moved by tides, currents, and winds on the water's surface.



△ Drifting
Some microscopic plankton can swim, but most float freely in the water and are carried along by ocean currents.



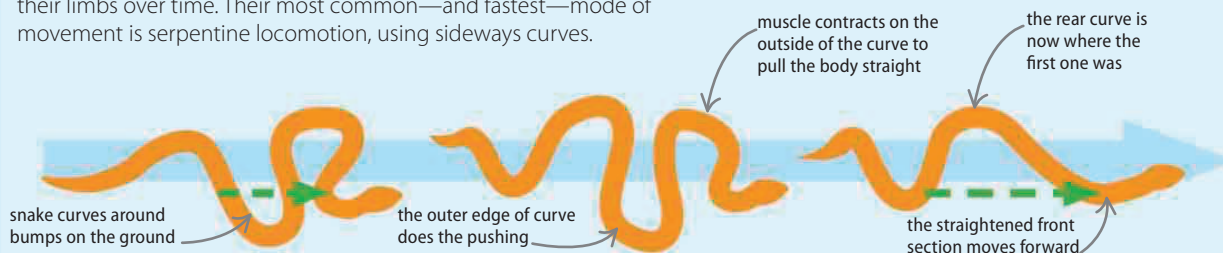
△ Swimming
Aquatic animals that can swim strongly enough to control where they move in the water are called nektons.



△ Staying still
Some organisms spend their lives anchored in one spot, usually under water, and just move their limbs to catch food.

Snake locomotion

Snakes evolved from four-legged reptiles, with their ancestors losing their limbs over time. Their most common—and fastest—mode of movement is serpentine locomotion, using sideways curves.



△ 1. Bunching up
The body is pulled into wide curves so the rear end moves toward the head.

△ 2. Stretching out
As the body straightens, the curved sections push against the rough ground.

△ 3. Gaining ground
The head gains ground by moving forward, and then the sequence starts again.